Coherent Phase Control in e-Ar Scattering in a Bichromatic Laser Field in the Second Born Approximation *

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We study the elastic scattering of atomic argon by an electron in the presence of a bichromatic laser field in the second Born approximation. The target atom is approximated by a simple screening potential. We explore the dependences of the differential cross sections on the relative phase φ between the two components of the radiation field and discuss the influence of the number of photons exchanged on the phase-dependence effect. Moreover, for different scattering angles and incident electron energies, the differential cross sections are notably different.

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The scattering of electrons by an atomic system in the presence of a laser field has attracted much attention since the pioneering work of Bunkin and Fedorov.[1] Summaries of these investigations can be found in the book by Mittleman,[2] the survey of Joachain,[3] the recent review.[4] In most of the theoretical work presented in this domain, the laser radiation has been treated as a classical radiation filed with a single frequency ω, or some narrow band multi-mode approximation has been employed, yielding better agreement with the experiments by Weigngarshofer.[5] As the experimental technology improved, powerful lasers and new kinds of laser fields have been applied to laser-assisted atomic and molecular processes, especially the multicolor laser. Therefore, the treatment of electronic collision processes in external fields, and in particular in laser fields is fundamentally important and is also of interest for some applications. The study of free-free transitions in a powerful bichromatic laser field has become feasible experimentally to coherently control the phase φ between the two components of the radiation field. This possibility stimulated a considerable body of research work on the coherent phase control of multiphoton processes.[6−18] The scattering in the most of the considerable body of research work in the bichromaitc laser field is treated in the first Born approximation (FBA). As an improvement of the theoretical treatment, in this Letter, we consider the elastic scattering of an electron by argon atom in the presence of an intense bicromaic laser filed in the second Born approximation (SBA), accompanied by transfer l photons. Atomic units $\hbar = m = e = 1$ are used throughout.

The laser field is described as a classical electromagnetic field[19] with the frequencies ω and 2ω, respectively. We write for the bichromatic laser field $\boldsymbol{\mathcal{E}}(t) = \boldsymbol{\mathcal{E}}_0[\sin\omega t + \sin(2\omega t + \varphi)]$, where $\boldsymbol{\mathcal{E}}_0$ is the electric field amplitude vector and the relative phase φ can be arbitrarily changed.

The target atom is described by a potential[20]

$$V(r) = -\frac{Z}{r}\sum_{i=1}^{3} A_i \exp(-\alpha_i r), \tag{1}$$

where r denotes the position of the electron with respect to the nucleus, and Z the nuclear charge number. For argon, $A_1 = 2.1912$, $A_2 = -2.8252$, $A_3 = 1 - A_1 - A_2$, $\alpha_1 = 5.5470$, $\alpha_2 = 4.5687$, and $\alpha_3 = 2.0446$.

The scattering matrix for the laser-assisted free-free transition in the second Born approximation reads

$$S_{fi}^{\langle 2\rangle} = S_{fi}^{(1)} + S_{fi}^{(2)}, \tag{2}$$

where

$$S_{fi}^{(1)} = -i\langle \chi_{\boldsymbol{k}_f}|V|\chi_{\boldsymbol{k}_i}\rangle, \tag{3}$$

where $\chi_{\boldsymbol{k}_f}$ and $\chi_{\boldsymbol{k}_i}$ are the initial and the final states of the electron, described by the Volkov wave function

$$\begin{aligned}\chi_{\boldsymbol{k}_{i,f}} = {} & \exp(i\boldsymbol{k}_{i,f}\cdot r) \\ & \times \exp\left[-iE_{i,f}t - \frac{i}{\omega^2}\boldsymbol{k}_{i,f}\cdot\boldsymbol{\mathcal{E}}_0\sin\omega t\right] \\ & \times \exp\left[-\frac{i}{4\omega^2}\boldsymbol{k}_{i,f}\cdot\boldsymbol{\mathcal{E}}_0\sin(2\omega t+\varphi)\right],\end{aligned} \tag{4}$$

where $\boldsymbol{k}_{i,f}$ are the wave vector of electron in the initial and final states and $E_{i,f}$ is the kinetic energies of the incident and ejected electron.

With using the potential of Eq. (1) and the wave functions like Eq. (4), The first term of the $S_{fi}^{\langle 2\rangle}$ matrix is then cast as

$$S_{fi}^{(1)} = -2\pi i\sum_l T_{f,i}^{(1)}(l)\delta(E_f - E_i + l\omega). \tag{5}$$

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The transition matrix element, resolved with respect to multiphoton exchange processes, has the form

$$T_{f,i}^{(1)}(l) = B_l\Big(\lambda, \frac{1}{4}\lambda, \varphi\Big)V(k_{f,i}), \tag{6}$$

in which

$$V(k_{f,i}) = \int d\boldsymbol{r} e^{-i(\boldsymbol{k}_f - \boldsymbol{k}_i)\cdot\boldsymbol{r}} V(r), \tag{7}$$

is the Fourier transformation of the potential, and

$$B_l\Big(\lambda, \frac{1}{4}\lambda, \varphi\Big) = \sum_{n=-\infty}^{\infty} J_{l-2n}(\lambda) J_n\Big(\frac{1}{4}\lambda\Big)\exp(-in\varphi), \tag{8}$$

is the generalized Bessel function with $\lambda = (\boldsymbol{k}_f - \boldsymbol{k}_i)\cdot \boldsymbol{\mathcal{E}}_0/\omega^2$. $J_n(\lambda)$ is the ordinary Bessel function.

The second term of the $S_{fi}^{\langle 2\rangle}$ matrix is then cast as

$$\begin{aligned} S_{fi}^{(2)} &= -i\int_{-\infty}^{\infty} dt \int d\boldsymbol{r} \int d\boldsymbol{r}' \int_{-\infty}^{\infty} dt' \chi_{\boldsymbol{k}_f}^*(\boldsymbol{r}, t) \\ &\quad \times V(r) G(\boldsymbol{r}, t; \boldsymbol{r}', t') V(r') \chi_{\boldsymbol{k}_i}(\boldsymbol{r}', t) \\ &= -2\pi i T_{fi}^{(2)}(l)\delta(E_f - E_i + l\omega), \end{aligned} \tag{9}$$

Where $G(\boldsymbol{r}, t; \boldsymbol{r}', t')$ is the Green function

$$G(\boldsymbol{r}, t; \boldsymbol{r}', t') = -\frac{i}{(2\pi)^3}\int d\boldsymbol{k} \chi_{\boldsymbol{k}}(\boldsymbol{r}, t)\chi_{\boldsymbol{k}}^*(\boldsymbol{r}, t) u(t - t'), \tag{10}$$

in which $u(x)$ is the step function.

With using the Green function of Eq. (10) and the wave functions of Eq. (4), we obtain

$$S_{fi}^{(2)} = -2\pi i T_{fi}^{(2)}(l)\delta(E_f - E_i + l\omega), \tag{11}$$

where the photon-number-resolved $T_{f,i}^{(2)}(l)$-matrix elements are given by the expression

$$\begin{aligned} T_{f,i}^{(2)}(l) &= \frac{1}{(2\pi)^3}\sum_m \int d\boldsymbol{k} \frac{1}{E_i - E - m\omega + i0^+} \\ &\quad \times V(\boldsymbol{k}_f, \boldsymbol{k}) V(\boldsymbol{k}, \boldsymbol{k}_i) B_m\Big(\lambda_1, \frac{1}{4}\lambda_1, \varphi\Big) \\ &\quad \times B_{l-m}\Big(\lambda_2, \frac{1}{4}\lambda_2, \varphi\Big). \end{aligned} \tag{12}$$

Thus we can obtain the scattering matrix $T_{f,i}^{\langle 2\rangle}$ with exchange of l photons in the second Born approximation

$$T_{f,i}^{\langle 2\rangle}(l) = T_{f,i}^{(1)}(l) + T_{f,i}^{(2)}(l). \tag{13}$$

The differential cross sections (DCS) for the net exchange of l photons between the colliding system and the bichromatic laser field in the second Born approximation is

$$\frac{d\sigma}{d\Omega} = \Big(\frac{1}{2\pi}\Big)^2 \frac{\boldsymbol{k}_f}{\boldsymbol{k}_i} |T_{f,i}^{\langle 2\rangle}(l)|^2. \tag{14}$$

For the numerical calculation, we study the relation between the DCS for electron-argon atom scattering and the relative phase angle φ under the scattering geometry of the experiment of Weingartshofer[5] in which the angle between the polarization vector $\boldsymbol{\mathcal{E}}$ and the momentum $\boldsymbol{k}_i$ of the incident electron is $\psi_0 = 38°$, the momentum $\boldsymbol{k}_f$ of the scattered electron is in the plane defined by the polarization vector $\boldsymbol{\mathcal{E}}$ and the momentum $\boldsymbol{k}_i$ of the incident electron. The bichromatic laser parameters are $\omega = 0.117\,\text{eV}$ and $\mathcal{E}_0 = 2.7\times 10^8\,\text{V}\cdot\text{cm}^{-1}$, all taken from Ref. [6].

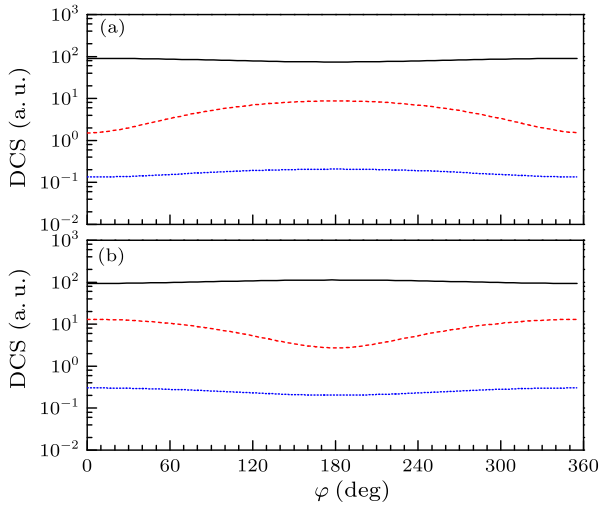

Fig. 1. (a) The DCS for elastic-argon scattering with absorption of one photon ($l = 1$), two photons ($l = 2$) and three photons ($l = 3$). The scattering angle of incident electron is $\theta = 13°$. The kinetic energies of the incident electron is E_i=9.5 eV. Solid curve: $l = 1$; dashed curve: $l = 2$; dotted curve: $l = 3$. (b) The same as (a). Solid curve: $l = -1$; dashed curve: $l = -2$; and dotted curve: $l = -3$.

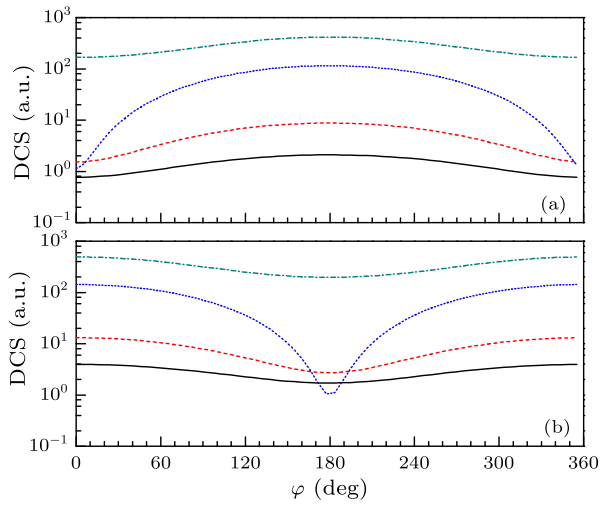

Fig. 2. (a) The DCS for the elastic electron-argon scattering with absorption of two photon ($l = 2$) at four scattering angles $\theta = 8°$ (solid curve), 13° (dashed curve), 30° (dotted curve) and 50° (dashed and dotted curve), respectively. The incident electron energy is $E_i = 9.5\,\text{eV}$. (b) The same as (a) but for $l = -2$.

Figure 1 shows the relation between the DCS and the phase angel φ for the different number of exchanged photons. The phase angel φ varies from 0° to 360°. All the curves are symmetric for $\varphi = 180°$.

In Fig. 1(a), the result for absorption is presented for $l = 1$, 2 and 3. In Fig. 1(b), the result for emission is presented for $l = -1$, -2 and -3. With all this, the results show that the phase-dependence effect for $l = \pm 2$ is obviously greater than that curves for $l = \pm 1$ and ± 3. The DCSs for $l = 2$, 3 and -1 are the largest at $\varphi = 180°$ and while those for $l = 1$, $l = -2$ and $l = -3$ reach their minima. This may attribute to the fact that the generalized Bessel function satisfies the relation[4]

$$B_{-l}(a, b, \varphi) = (-1)^l B_l^*(a, b, \varphi - \pi). \tag{15}$$

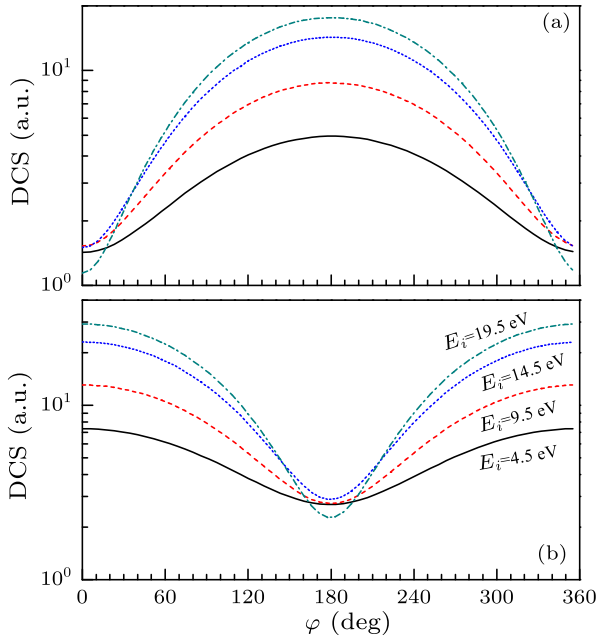

Fig. 3. (a) The DCS for the elastic electron-argon scattering with absorption of two photons ($l = \pm 2$), with four incident electron energies $E_i = 4.5$ eV (solid curve), 9.5 eV (dashed curve), 14.5 eV (dotted curve) and 19.5 eV (dashed and dotted curve), respectively. The scattering angle is $\theta = 13°$. (b) The same as (a) but for $l = -2$.

Figure 2 shows the results for $l = \pm 2$ at the scattering angles $\theta = 8°$, 13°, 30°, and 50°. The DCS at $\theta = 13°$ is more sensitive to φ than at other angles. At this angle, the result for bremsstrahlung is maximally enhanced at $\varphi = 180°$. In this case, one can effectively control the dynamics by varying the phase.

Figure 3 displays the result of $l = \pm 2$ for different impact energies at the scattering angle $\theta = 13°$ where the phase dependence is most sensitive. When the impact energies increases, the result becomes more and more sensitive to the variation of phase. The higher impact energy, the scattered electron may exchange more photons with the laser field, while the results for $l = \pm 2$ only make a small contribution.

In summary, the electron-argon elastic scattering in a bichromatic laser field is investigated for a simple potential in the second Born approximation. The dependence of DCS on the relative phase φ is investigated at different scattering angles and impact energies. These are important in the coherent control in the free-free transition.

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